LEX AI

CONFIDENTIAL · 2026

DATA INFRASTRUCTURE & COMPUTE PROFILE

Ukrainian Legal Intelligence Platform

legal.org.ua · SecondLayer

legal.org.ua

132M	55M	570B	12.3TB
COURT DECISIONS	REGISTRY RECORDS	EMBEDDING TOKENS	VECTOR STORE

1 Data Infrastructure Overview

3 databases · ~152 GB active · 12.6 TB target

132M	8.87M	30.6M	50K
COURT DECISIONS NATIONAL REGISTRY	BUSINESS ENTITIES ЄДР + ФОП	ENFORCEMENT PROCEEDINGS	LAWS & CODES LEGISLATION

⚖ MCP_BACKEND — COURT & LEGISLATION

Court Decisions	166,972 ✓
Doc Sections	466,126
Legislation Arts	5,202
Target (full)	132,135,013

PostgreSQL :5432 · 2 GB active · → 6.5 TB target

🏛 MCP_OPENREYESTR — STATE REGISTRIES

Enforcement Proc.	30,605,224
Debtors Registry	13,325,901
ФОП (Entrepreneurs)	6,888,292
Legal Entities ЄДР	1,986,542
Legal Acts / Other	2,235,345

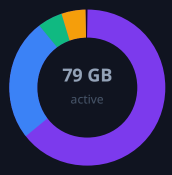
PostgreSQL :5435 · 79 GB active · ~95% complete

🏛 MCP_RADA — PARLIAMENT DATA

Bills (skl9)	14,650
Voting Records	5,026
Deputies / Assistants	4,859

PostgreSQL :5433 · 2 GB active · Full text: 0% (todo)

STORAGE DISTRIBUTION (ACTIVE)



Enforcement	51 GB	65%
Debtors	20 GB	25%
FOP Registry	3.9 GB	5%
Other	4.1 GB	5%

UPDATE FREQUENCIES

SOURCE	FREQ
Court Decisions	ON-DEMAND
ЄДР / ФОП	DAILY XML
Enforcement	DAILY CSV
Debtors	DAILY CSV
Bankruptcy	DAILY XML
RADA Bills	DAILY API
RADA Voting	DAILY API
Notaries / Experts	WEEKLY XML
Legislation	MANUAL SYNC

COURT DECISION PROFILE

Avg length	21,073 chars
Avg tokens	5,268 tokens
Sections / doc	19 avg
Section size	485 chars / 121 tok
Dominant type	LAW_REFS (53%)

2 Document Processing Pipeline

5-stage · parse → section → embed → pattern → store

Parse	Sectionize	Embed	Patterns	Store
PDF · DOCX · HTML TXT · RTF	Regex → 6 types LLM fallback 10%	BGE-M3 / voyage 1024-dim · 512 tok	Legal pattern store ≥3 cases · cos≥0.7	PostgreSQL + Qdrant 2.64B vectors total
CPU	CPU 90%	GPU ★ CRITICAL	CPU	CPU/IO
OCR fallback	GPU 10%			

SECTION TYPE DISTRIBUTION

LAW_REFERENCES	53%
DECISION	23%
FACTS	14%
CLAIMS	4%
COURT_REASONING	4%
AMOUNTS	2%

🧠 EMBEDDING TOKEN BREAKDOWN — 570.8B TOTAL

Court Decisions · 568.0B (99.5%)

Legislation 0.95B · OpenReyestr 1.77B

SOURCE	TOKENS	VECTORS
Court Decisions (132M)	568.0B	2.64B
Legislation (50K laws)	0.95B	1.85M
OpenReyestr entities	1.77B	8.8M
TOTAL	**570.8B**	**2.65B**

Qdrant storage: 1024-dim float32 = 12.3 TB total vector index

CPU PROCESSING CAPACITY

TASK	1 CORE	20 THREADS
XML/CSV Import (OpenReyestr)	20 min	20 min
Regex Sectionizing (132M docs)	1,835 h	4 days
DB Inserts (2.64B sections)	14 h	14 h
Qdrant Indexing	3 days	3 days

Required: 16–32 CPU cores · standard server sufficient

⚡ LLM INFERENCE — BULK PROCESSING: 30.3B TOKENS

TASK	DOCS	TOKENS
Sectionizer fallback (10%)	13.2M	29.1B
Beneficiary chain analysis	773K	1.16B
Real-time (1K users/day)	15K req	75M/day
Bulk total	–	**30.3B**

Note: court decision metadata extracted via regex — LLM only for complex fallback cases

3 GPU Compute Analysis

570.8B embedding tokens · 30.3B LLM tokens

HARDWARE COMPARISON — BULK PROCESSING TIMELINE

1× H100 80GB	4× H100 (emb) + API LLM	4× H100 (emb) + 4× H100 (LLM)	4× A100 80GB + API LLM	8× RTX 4090 + API LLM
66 days embedding	17 days embedding ✅	17 days embedding ✅	27 days embedding ✅	66 days embedding ⚠️
720 days LLM (bottleneck)	$30K LLM via API	35 days LLM parallel ✅	$30K LLM via API	$30K LLM via API
	RECOMMENDED			

EMBEDDING THROUGHPUT (TOKENS/SEC)

4× H100	400K/s
4× A100	240K/s
2× H100	200K/s
8× RTX 4090	100K/s
1× H100	100K/s

💰 COST: BULK PROCESSING (ONE-TIME)

All via Cloud API VoyageAI $0.02/1M + OpenAI $0.15–0.60/1M	$30,087
Embed API + Self-host LLM VoyageAI + Qwen 72B on 4× H100 (35 days electricity)	$13,415
Cloud rental 8× H100 · 35 days ★ 8 GPUs × 35d × 24h × $2.50/h — then buy 4× H100	$7,200
Self-hosted 8× H100 (capex) $200K hardware, 5+ years, ~$7K electricity/year	$200K

STORAGE TARGETS: ACTIVE → FULL DATASET

COMPONENT	ACTIVE NOW	FULL TARGET
PostgreSQL (Backend)	2 GB	6.5 TB
PostgreSQL (OpenReyestr)	79 GB	~160 GB
PostgreSQL (RADA)	2 GB	~45 GB
Qdrant (Court vectors)	<1 GB	10.8 TB
Qdrant (Other vectors)	<1 GB	1.5 TB
TOTAL	~83 GB	~18.8 TB

ONGOING COMPUTE (1K USERS · DAILY)

WORKLOAD	VOLUME/DAY
Real-time LLM inference	~75M tokens
New court decisions	~5K docs → embed
OpenReyestr delta updates	~50K records
RADA bills / voting	~200 records
GPU Needed	4× H100 sufficient

MODEL SELECTION

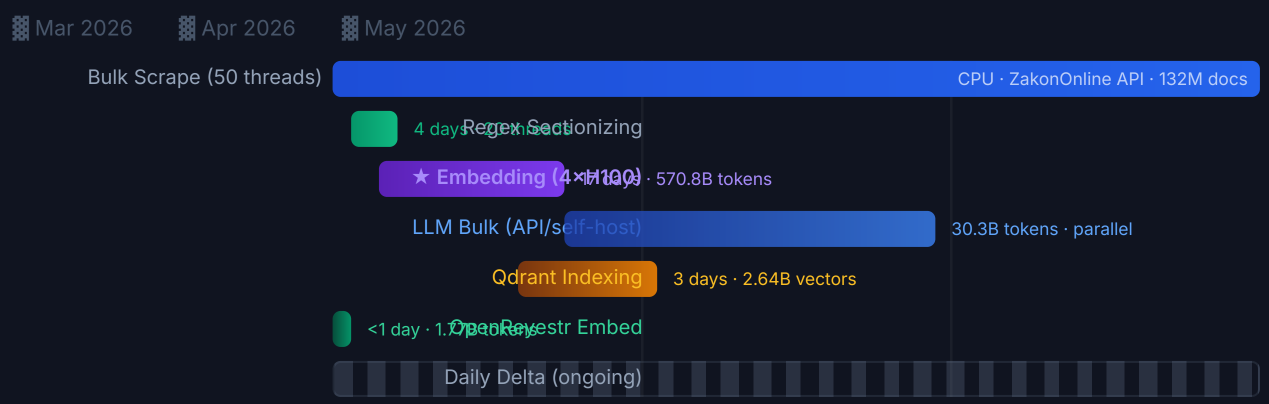
TASK	MODEL
Inference (chat)	QWEN 2.5 72B
Inference (fallback)	LLAMA 3.3 70B
Embeddings	BGE-M3 (1024-DIM)
OCR (doc parsing)	GOOGLE VISION API

4

3-Month Bulk Processing Roadmap

Target: 132M court decisions fully indexed by May 2026

EXECUTION TIMELINE



Month 1 — March 2026: Data Acquisition

- Launch 50-thread bulk scraper for reyestr.court.gov.ua
- Target: 1M docs/day from ZakonOnline API
- Regex sectionizing pipeline: 20 threads → 4 days
- OpenReyestr: complete debtors (+3M) & enforcement (+1.8M) delta
- Start embedding pipeline on first 20M completed docs

Month 2 — April 2026: Embedding Blitz

- 4× H100 dedicated to BGE-M3 embedding @ 400K tok/s
- Complete 570.8B tokens in 17 days
- Qdrant: index 2.64B vectors, 10.8 TB storage
- LLM bulk: Together.ai API for 30.3B token sectionizer fallback
- RADA: load all 14,650 bill full texts (skl6–skl9)

Month 3 — May 2026: Enrichment & Launch

- LegalPatternStore: extract patterns from 132M decisions
- Beneficiary chains: 773K entity analysis via LLM
- Legislation: embed 50K laws (947M tokens, <1 day)
- QA: semantic search validation, citation accuracy testing
- Launch: scale from 3 → 50 → 1,000 attorney users

3-MONTH SUMMARY: BY THE NUMBERS

METRIC	TARGET
Court decisions indexed	132,135,013
Semantic sections stored	2,640,000,000
Qdrant vectors total	2,650,000,000
Embedding tokens processed	570.8B
LLM tokens (bulk API)	30.3B
PostgreSQL total storage	~6.7 TB
Qdrant vector storage	~12.3 TB
Total storage footprint	~18.8 TB

API USAGE ON TOGETHER.AI

PERIOD	REQ/DAY	TOKENS/DAY
Now (3 users)	50–100	250K
Month 1–3 (bulk)	batch	~337M/day
Post-launch (1K users)	15–20K	75M/day

Primary models: Qwen 2.5 72B Instruct · Llama 3.3 70B Instruct
Embeddings: multilingual model (replacing VoyageAI)